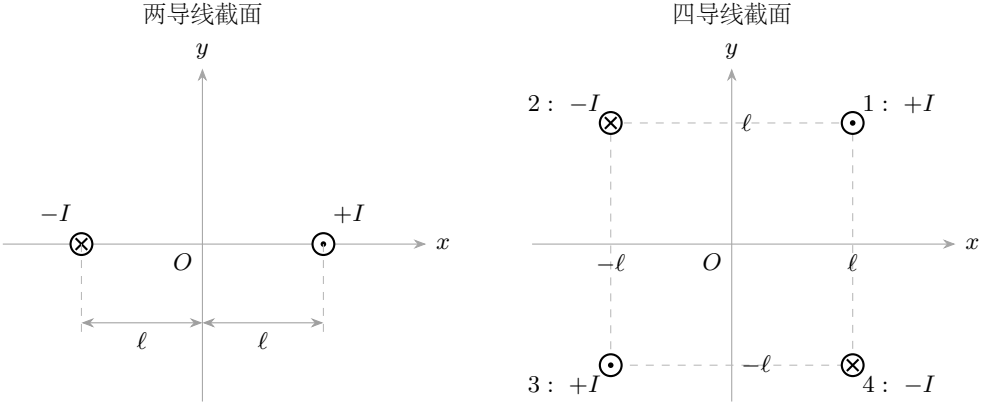
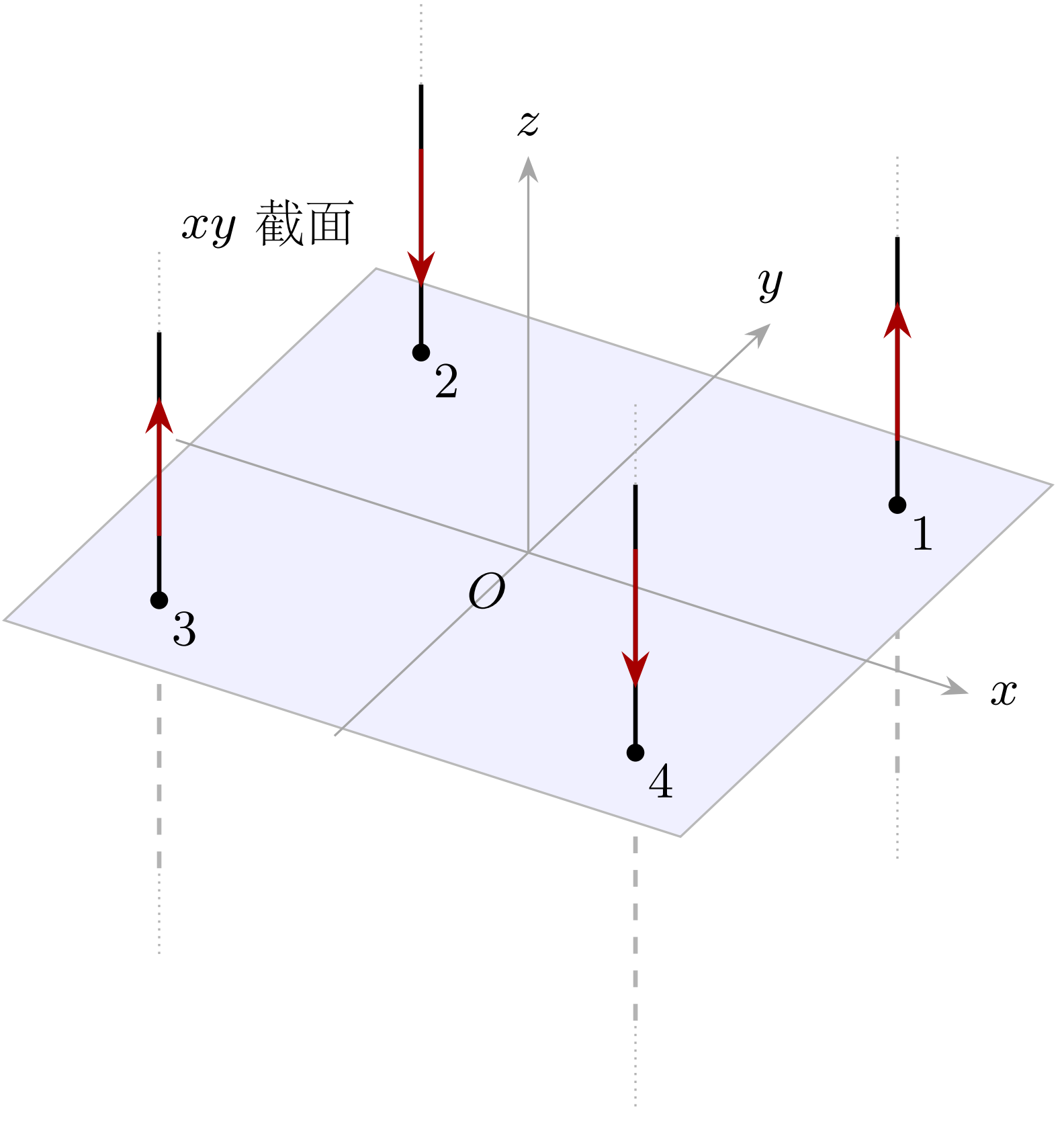
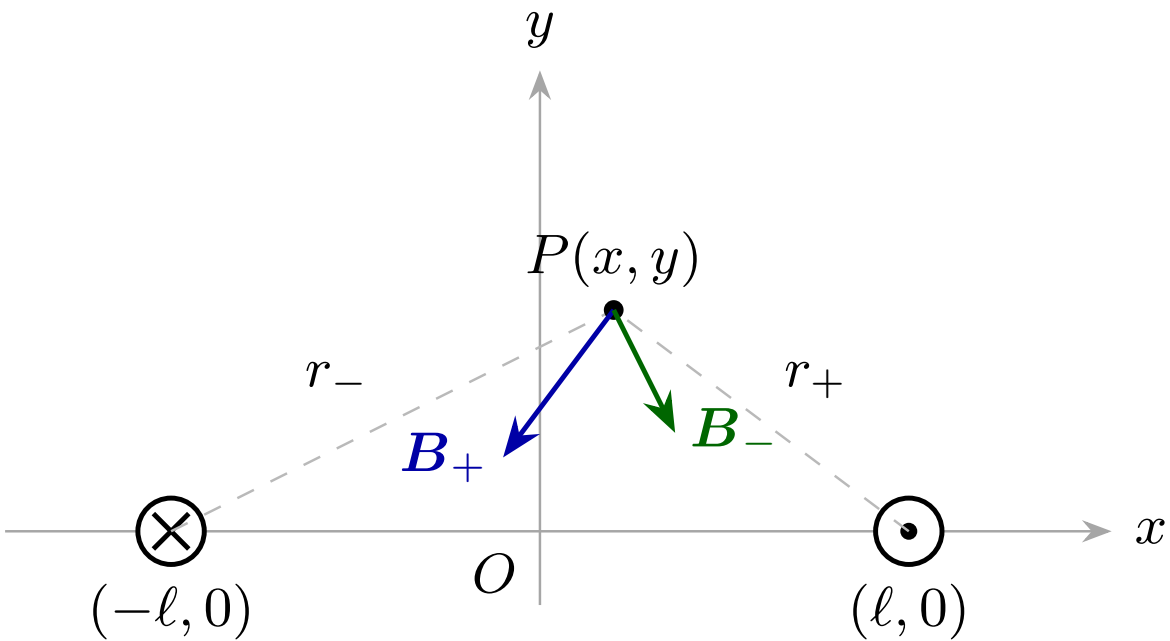
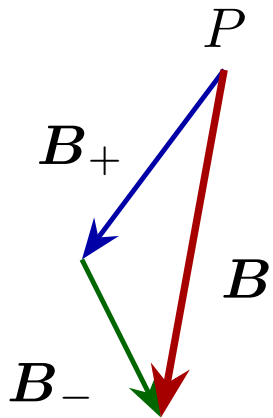
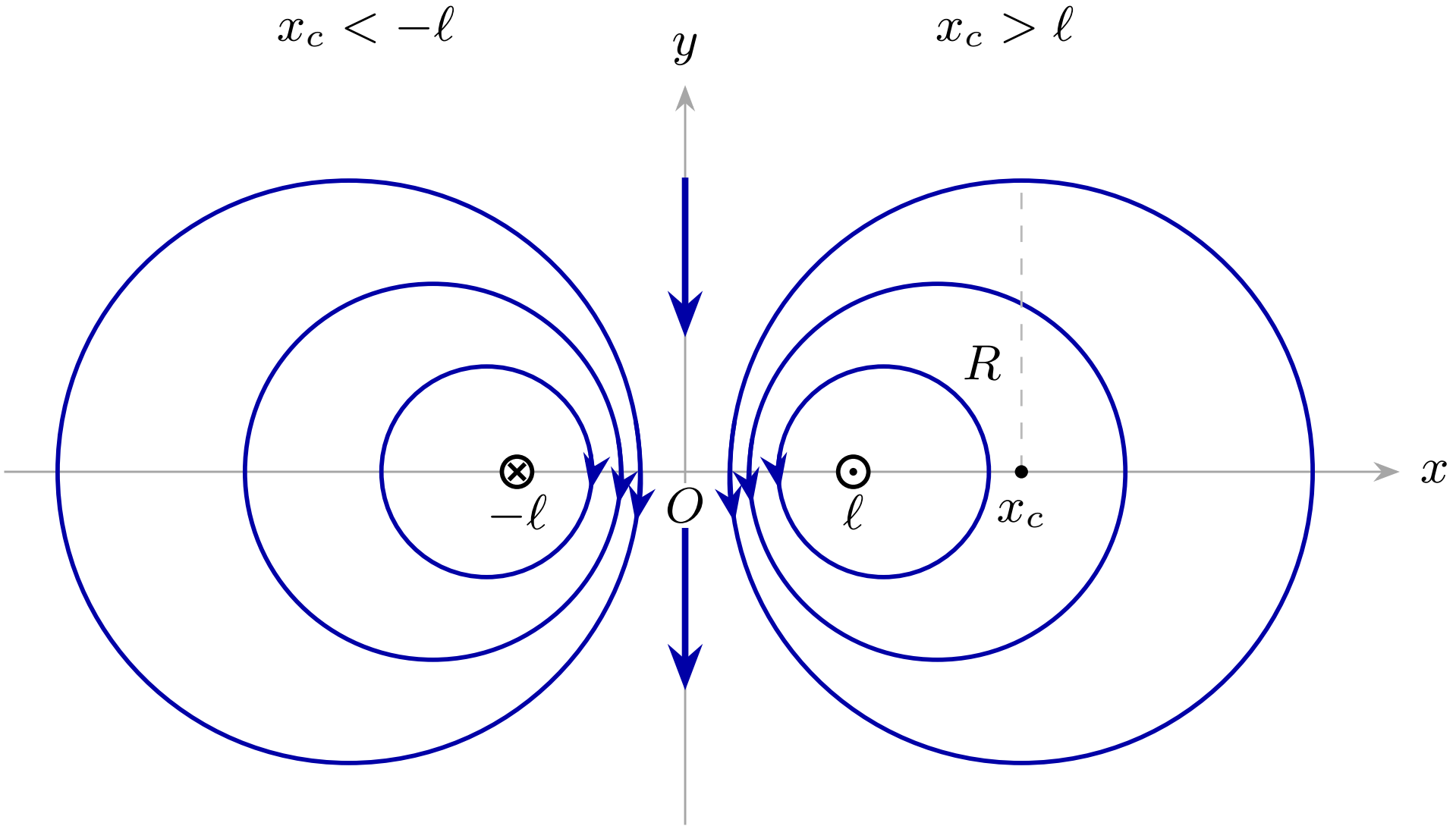

两导线截面
y
x
O
−I
+I
ℓ
ℓ
四导线截面
y
x
O
2： −I
1： +I
3： +I
4： −I
ℓ
−ℓ
−ℓ
ℓ
⊙： +z,
⊗： −z


xy 截面
z
y
x
O
1
2
3
4

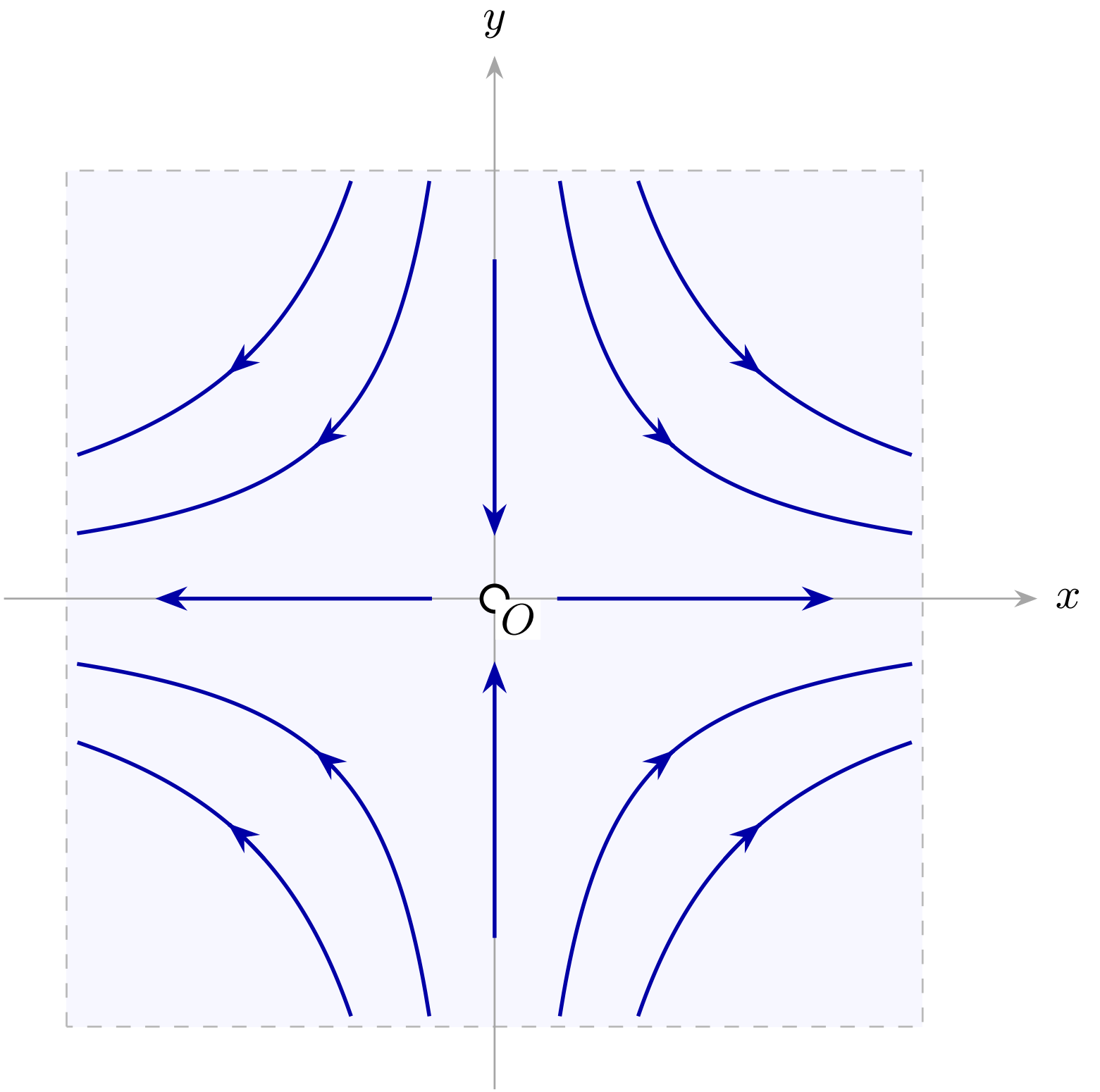

y
$P(x,y)$
r_-
r_+
$\boldsymbol{B}_+$
$\boldsymbol{B}_-$
x
O
$(-\ell,0)$
$(\ell,0)$

首尾相接


P
$\boldsymbol{B}_+$
$\boldsymbol{B}$
$\boldsymbol{B}_-$


$x_c < -\ell$
$x_c > \ell$
y
R
x
$-\ell$
O
ℓ
x_c

$$\boldsymbol{B} \simeq q(x, -y), \quad q > 0$$



局部窗口：$|x|, |y| \ll \ell$